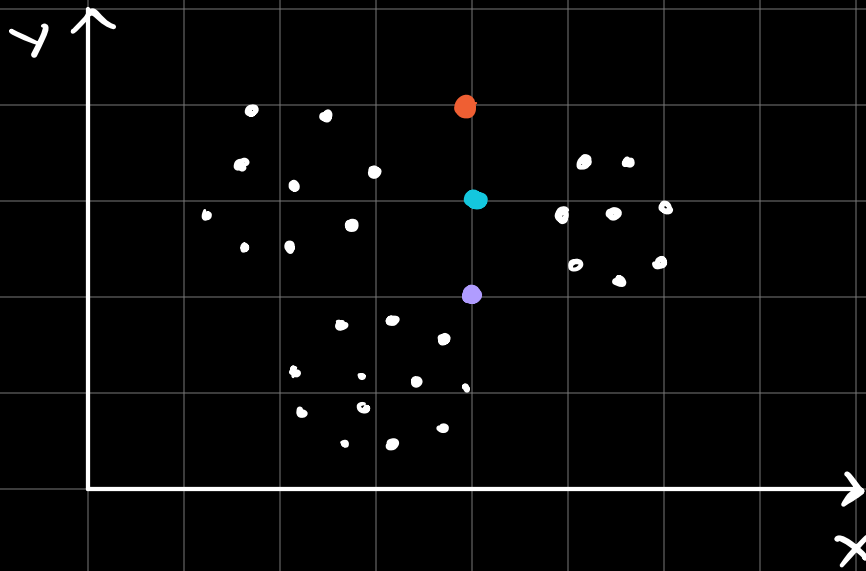


K-MEANS



$k \rightarrow$ no. of clusters

● $\rightarrow$ CENTROID 1

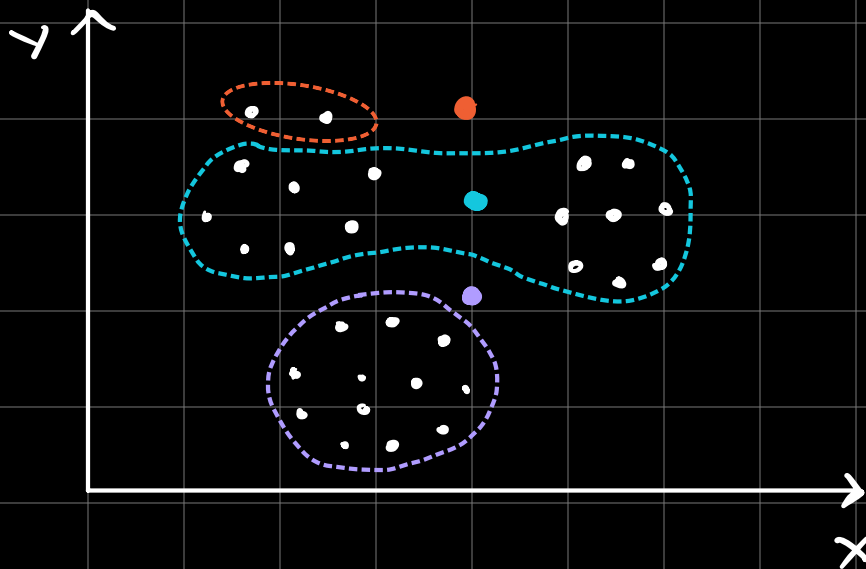
● $\rightarrow$ CENTROID 2

● $\rightarrow$ CENTROID 3

Step 1) Choose k centroids and initialize them.

Step 2) Calculate the distance of each centroid from each point.

Step 3) Assign the point a class the same as the closest centroid.



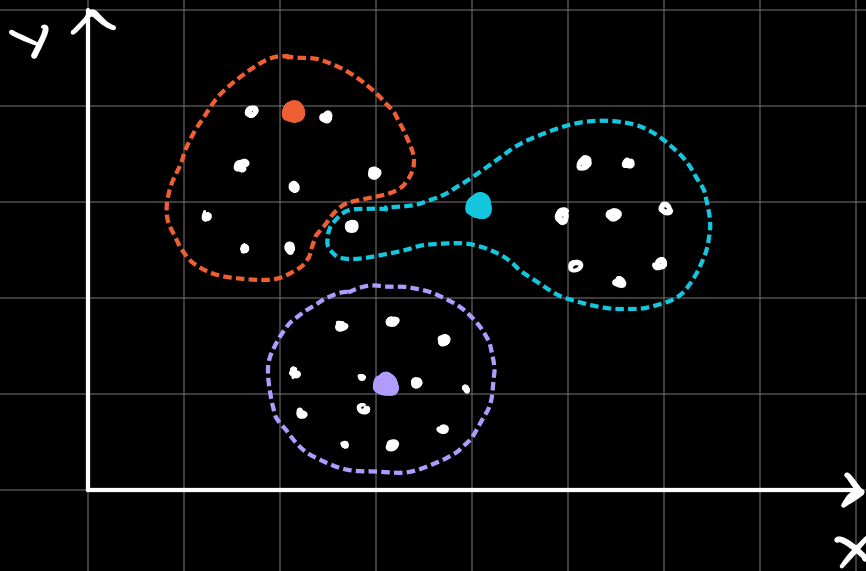
$k \rightarrow$ no. of clusters

● $\rightarrow$ CENTROID 1

● $\rightarrow$ CENTROID 2

● $\rightarrow$ CENTROID 3

Step 4) Find the mean distance b/w all the points and place the centroid of that class at that position.



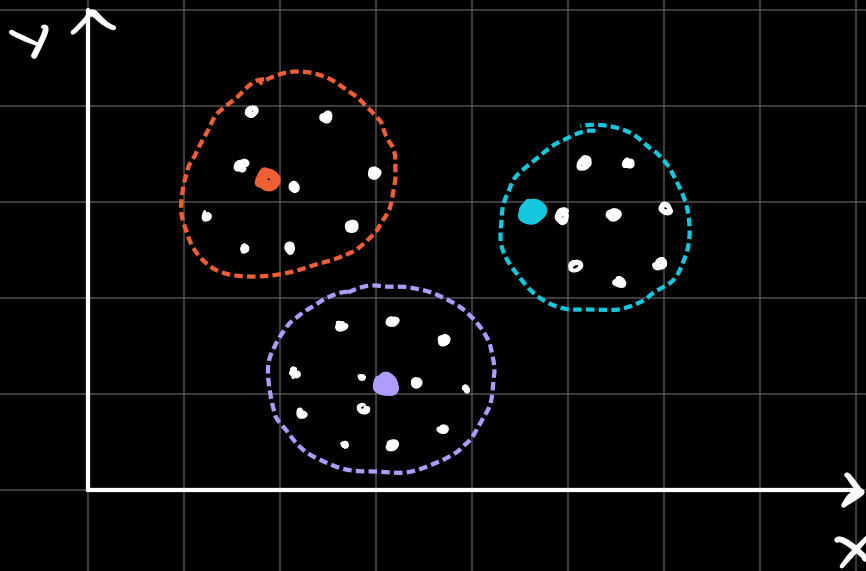
$k \rightarrow$ no. of clusters

● $\rightarrow$ CENTROID 1

● $\rightarrow$ CENTROID 2

● $\rightarrow$ CENTROID 3

Step 5) Repeat the steps till the position of the centroid stops changing.

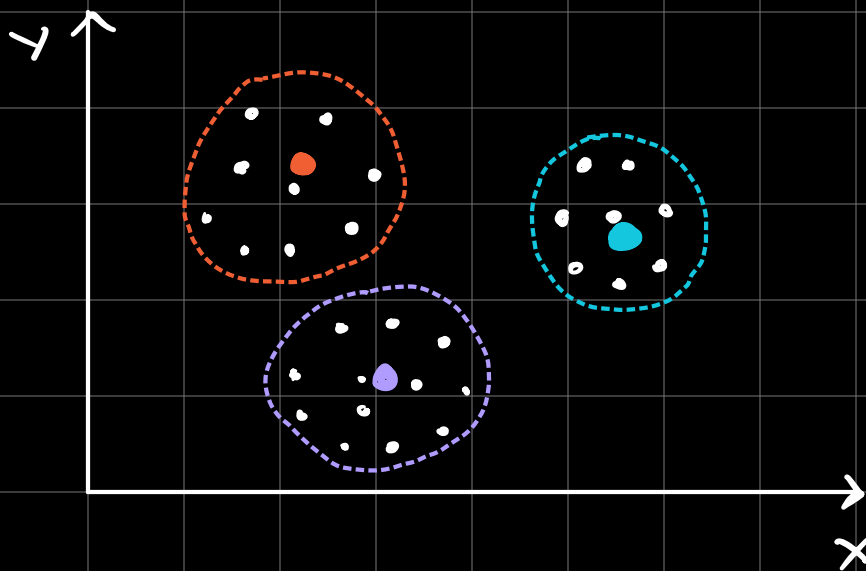


$k \rightarrow$ no. of clusters

● $\rightarrow$ CENTROID 1

● $\rightarrow$ CENTROID 2

● $\rightarrow$ CENTROID 3



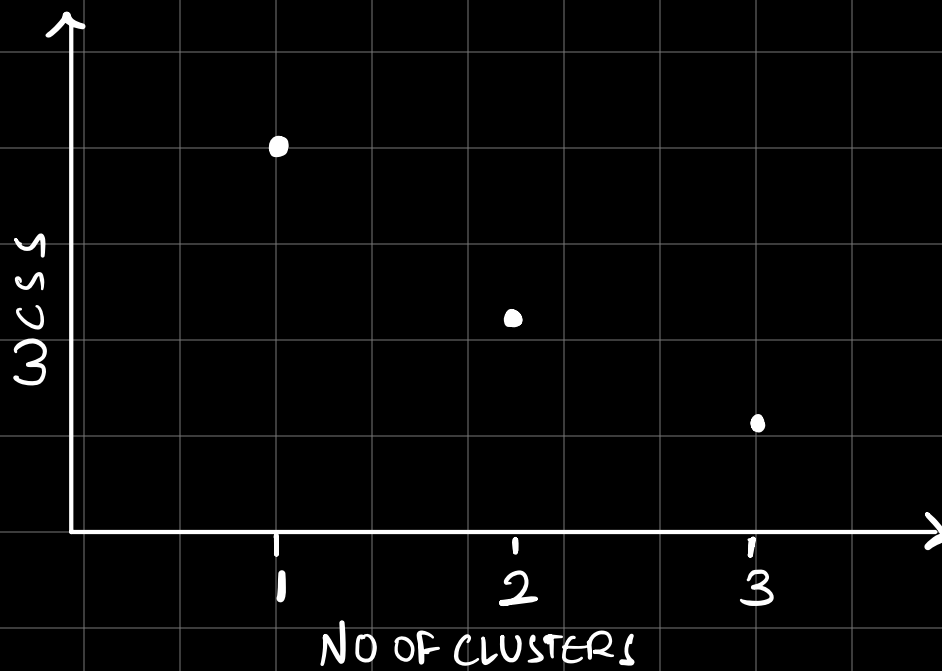
$k \rightarrow$ no. of clusters

● $\rightarrow$ CENTROID 1

● $\rightarrow$ CENTROID 2

● $\rightarrow$ CENTROID 3

Elbow Method:



WCSS (INERTIA) $\rightarrow$ WITHIN CLUSTER SUM OF SQUARED DISTANCES

$$\underline{\text{NET WCSS}} = \sum_{i=1}^n \text{WCSS}_i$$

From the graph,

$$\text{WCSS}_1 > \text{WCSS}_2 > \text{WCSS}_3 < \dots < \text{WCSS}_n$$

Elbow Point: Point where decrease of WCSS almost stabilizes